KrishiGrid Bangladesh
Data-Driven Food Security Simulation

KRISHIGRID BANGLADESH

A Data-Driven Simulation of Food Import Dependency and Decentralized Agricultural Infrastructure

PART 3 — THE CROP REVOLUTION

What should Bangladesh grow, where should it grow, and how much more can it produce?

Bangladesh already has the land and capacity, but smarter crop selection and placement can unlock significantly higher food and edible oil production.

Analysis : 19 crops, 64 districts, ~400 district crop combinations
Tools : SQL, Power BI
Dataset : crop_research.csv 19 rows 28 columns
district_crop_suitability.csv ~400 rows 15 columns

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THE CROP REVOLUTION

THE QUESTION

What should Bangladesh grow, where should it grow, and how much more can it produce?

This analysis reveals how smarter crop selection and placement can unlock Bangladesh's full agricultural potential.

APPROACH

- *Analyzed 19 major crops across Bangladesh*
- *Mapped crop suitability across 64 districts*
- *Evaluated yield potential and additional cultivable area*
- *Assessed oil content and production impact for oilseed crops*
- *Identified seasonal opportunities within the Aman Boro gap*
- *Linked crops with required processing and infrastructure needs*

DATASET OVERVIEW

File : crop_research.csv
Rows : 19
Columns: 28

File : district_crop_suitability.csv
Rows : ~400
Columns: 15

KEY FIELDS

* Crop yield ton per hectare
* Additional cultivable area hectare
* Oil content percent
* Seasonal fit Aman Boro gap
* District suitability score
* Potential additional production MT
* Infrastructure requirement

SOURCE : FAO, BARI, DAE, Agricultural Research Data

SQL Queries

Question
Which crops offer the highest additional production potential based on yield and expandable land?

```
SELECT
   crop_name,

   ROUND(avg_yield_ton_per_hectare, 2) AS
yield_per_hectare,
   ROUND(potential_additional_area_hectare, 0) AS
additional_land,

   ROUND(
      avg_yield_ton_per_hectare *
potential_additional_area_hectare,
   0) AS potential_production_mt

FROM bdcrops

ORDER BY potential_production_mt DESC;
```

Key Insight:

Short duration Aman Rice and Mustard Rapeseed dominate total production potential at ~3.8 million MT and ~2.7 million MT respectively, driven by large expandable land.

High yield crops like Potato and Maize add significant output despite smaller land bases, showing the importance of productivity alongside scale.

A balanced strategy combining staple crops and high yield crops can maximize total food production without expanding agricultural land.

Question
Which oilseed crops have the strongest potential to reduce edible oil imports based on yield, oil content, and expansion capacity?

```
SELECT
   crop_name,

   ROUND(avg_yield_ton_per_hectare, 2) AS yield_per_hectare,
   ROUND(potential_additional_area_hectare, 0) AS additional_land,

   ROUND(
      avg_yield_ton_per_hectare * potential_additional_area_hectare,
   0) AS potential_production_mt,

   ROUND(oil_content_pct, 2) AS oil_content_pct,

   ROUND(
      (avg_yield_ton_per_hectare * potential_additional_area_hectare
* oil_content_pct) / 100.0,
   0) AS potential_oil_output_mt

FROM bdcrops

WHERE oil_content_pct > 0

ORDER BY potential_oil_output_mt DESC;
```

Key Insight:

Mustard Rapeseed dominates edible oil potential with over 1.08 million MT, far exceeding all other crops due to its large expansion capacity.

Sunflower and Groundnut provide strong secondary contributions at ~0.38M and ~0.32M MT, offering higher efficiency but lower scale.

This indicates that rapid import reduction should prioritize mustard expansion first, while gradually diversifying into higher oil yield crops for long term stability.

SQL Queries

Question
Which crops are best suited for different land types such as coastal, char, hilly, haor, barind, and plain regions?

```
SELECT
   crop_name,
   suitable_area_types

FROM bdcrops

ORDER BY crop_name;
```

Key Insight:

Most crops are highly concentrated in plain land, with many also adaptable to char and coastal regions, indicating where expansion is most immediately feasible.

In contrast, relatively few crops are suitable for hilly and haor areas, showing structural limitations in these regions.

This suggests that the fastest production gains will come from optimizing already cultivable land types rather than expanding into constrained terrains.

Question
Which districts have the highest suitability and expansion potential for oilseed cultivation?

```
SELECT
   d.district_name,
   d.division,
   c.crop_name,

   ROUND(d.potential_additional_area_hectare, 0) AS suitable_land

FROM bd_district_crop d
JOIN bdcrops c
   ON d.crop_id = c.crop_id

WHERE c.oil_content_pct > 0

ORDER BY suitable_land DESC;
```

Key Insight:

Oilseed expansion potential is heavily concentrated in Mustard Rapeseed, with districts like Jessore and Sirajganj offering up to ~**35K** hectares of additional cultivable land.

This shows that Bangladesh's edible oil strategy is geographically clustered, particularly in the Khulna and Rajshahi regions.

Targeting these high concentration districts can deliver rapid production gains, rather than spreading efforts across multiple lower potential crops.

Question
Which crops can be introduced to create a three crop cycle by utilizing seasonal gaps?

```
SELECT
   crop_name,

   ROUND(
      (CAST(duration_days_min AS INT) + CAST(duration_days_max AS INT)) / 2.0,
   0) AS avg_growth_days

FROM bdcrops

WHERE (CAST(duration_days_min AS INT) + CAST(duration_days_max AS INT)) / 2.0 <=
100

ORDER BY avg_growth_days ASC;
```

Key Insight:

Crops like Mung Bean (~70 days), Mustard (~85 days), and Potato (~85 days) offer the best opportunity to utilize short seasonal gaps between major crop cycles.

These short duration crops can significantly increase annual production without requiring additional land.

Expanding these crops is a practical pathway to achieving a three crop system and improving overall land productivity.

Question
If all suitable additional land is utilized, how much total crop and edible oil production could Bangladesh generate?

```
SELECT

   ROUND(SUM(potential_additional_production_mt), 0) AS
total_crop_production_mt,

   ROUND(SUM(estimated_oil_from_potential_mt), 0) AS
total_oil_production_mt

FROM bdcrops;
```

Key Insight:

Bangladesh can generate approximately **12.7 million** metric tons of additional crop production and around **2.0 million** metric tons of edible oil by utilizing existing untapped land.

This indicates a massive hidden production capacity within the current agricultural system without expanding total farmland.

Unlocking this potential can significantly reduce both food and edible oil import dependency in the short to medium term.

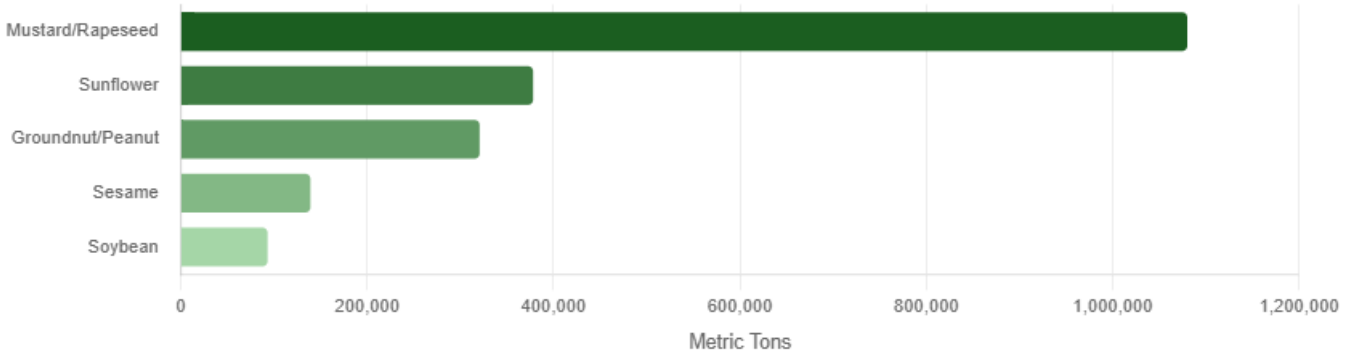
12.69
Million MT
Total Additional Crop Potential

2.01
Million MT
Total Oil Production Potential

Mustard/Rapeseed
Top Oilseed Crop

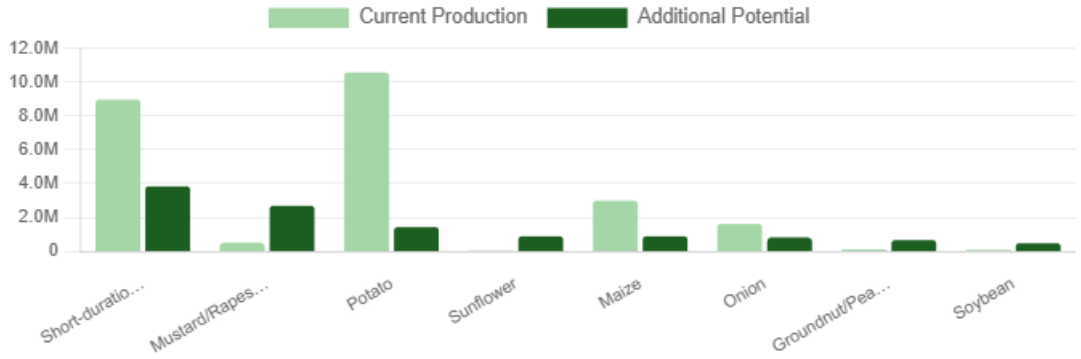
Mustard/Rapeseed
Best Gap-Planting Crop

Oilseed Crop Oil Production Potential (MT)



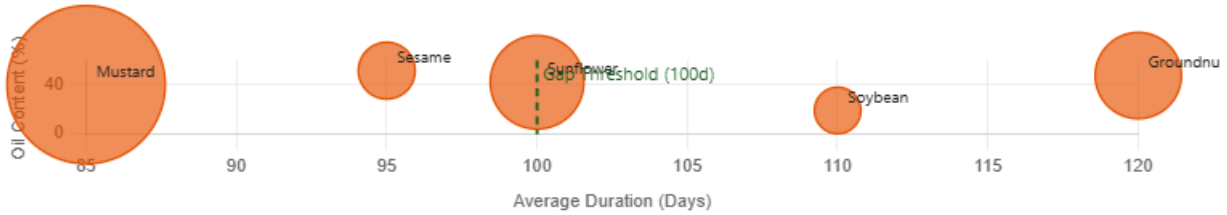
Mustard/Rapeseed dominates oil potential at 1.08M MT. Rapid import reduction should prioritize mustard first. Sunflower and groundnut are strong secondary contributors.

Current vs Additional Production Potential (Top 8 Crops)



Bangladesh has massive hidden production capacity from existing untapped land. No new land needed — just better cropping patterns and processing infrastructure.

Crop Duration vs Oil Content (%)



Crops below 100 days are ideal for gap planting between Aman and Boro seasons. Mustard combines short duration with high oil content.

Total Oil Potential Share by Oilseed Crop

Mustard alone accounts for over half of Bangladesh's total domestic oil production potential — KrishiGrid's oil mills create the biggest impact here.

KrishiGrid Bangladesh

District-Crop Suitability & Hub Equipment Mapping

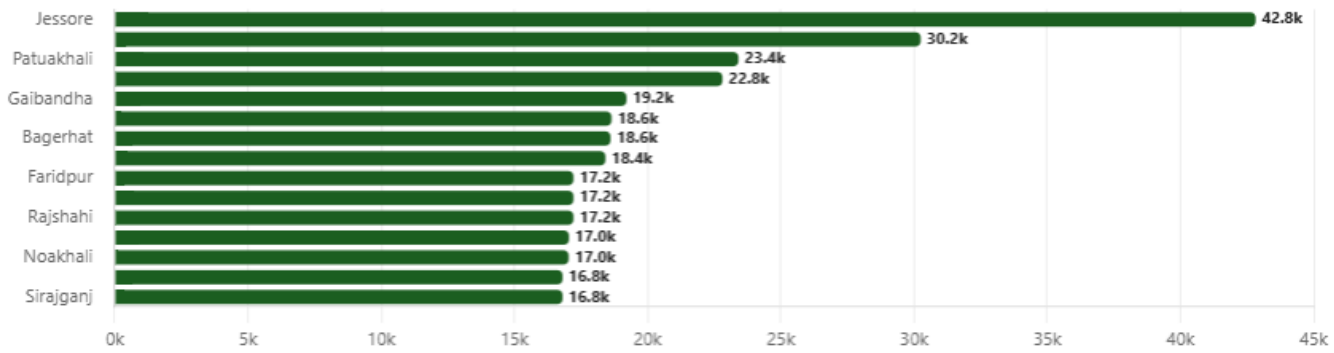
505
District-Crop Combinations
COMBINATIONS

483
Currently Grown Combinations
ACTIVE

283
High Priority Expansion Areas
HIGH PRIORITY

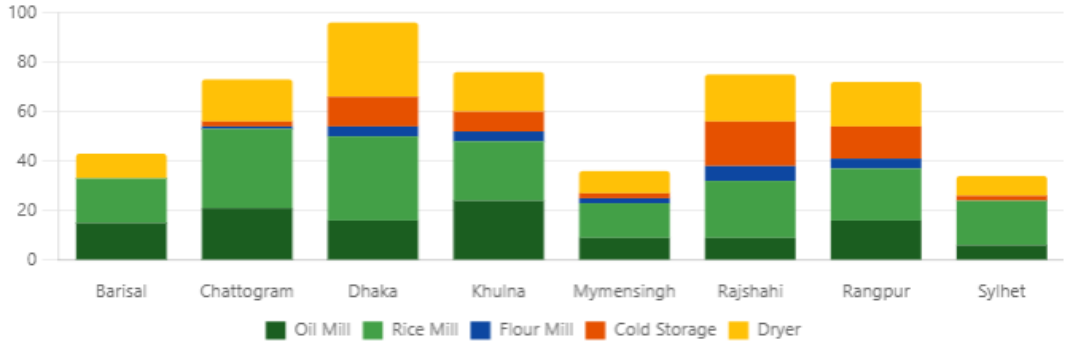
761,960
Total Oil Expansion Potential
METRIC TONNES

Top 15 Districts by Oilseed Expansion Potential (MT)



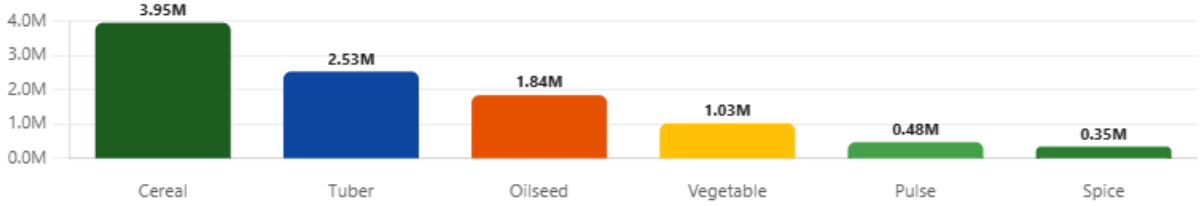
Oilseed expansion potential is geographically clustered in Khulna and Rajshahi regions. Targeting high-concentration districts delivers rapid gains versus spreading resources.

Equipment Required by Division



Equipment needs vary sharply by division. Oil mills dominate in southern divisions, cold storage in the north. Regional customization is not optional — it is essential.

Total Additional Production Potential by Crop Category



Oilseed crops have the highest expansion potential among all categories. Combined with oil mill ward hubs, this directly attacks the import dependency problem.

Hub Equipment Distribution — All Districts

Oil mills are the most needed piece of equipment across all district-crop combinations, confirming the central role of oil processing in the KrishiGrid design.

KEY INSIGHTS

Part 2/8

1 Bangladesh can generate ~12.7 million MT of additional crop production using existing untapped agricultural land.

2 Edible oil potential alone reaches ~2.0 million MT, with Mustard Rapeseed contributing over 1.08 million MT, making it the primary driver of import reduction.

3 Oilseed expansion is highly concentrated in specific districts such as Jessore and Sirajganj, showing that targeted regional investment will outperform broad national programs.

4 Most crops are best suited to plain land, with limited adaptability in hilly and haor regions, indicating that immediate gains lie in optimizing
already cultivable areas rather than expanding into constrained terrain.

5 Short duration crops like Mung Bean and Mustard enable additional crop cycles within existing farming seasons, unlocking higher annual output without increasing land use.

COMING NEXT

PART 4 of 8: THE INFRASTRUCTURE GAP

Production is not the real problem. In Part 4, I show how storage, processing, and supply chain inefficiencies prevent Bangladesh from turning production into real economic value.